CS5487: Machine Learning: Principles and Practice

Assignment 1

Semester A, 2026/27

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Total: 100 marks

- There are eight questions.
- Questions 1–4 are calculation problems worth 50 marks in total.
- Questions 5–8 are derivation problems worth 50 marks in total.
- Show the essential reasoning leading to each answer.
- Numerical answers without supporting work may receive limited credit.
- A calculator may be used ffor numerical evaluation.

Question	1	2	3	4	5	6	7	8	Total
Marks	10	10	10	20	10	10	10	20	100

Question 1: Gaussian Geometry [10 marks]

A two-dimensional observation is modeled as

$$\boldsymbol{X} \sim \mathcal{N}\left(\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}4 & 1\\1 & 1\end{bmatrix}\right).$$

Consider

$$\boldsymbol{x}_A = \begin{bmatrix}3\\3\end{bmatrix}, \qquad \boldsymbol{x}_B = \begin{bmatrix}1\\4\end{bmatrix}.$$

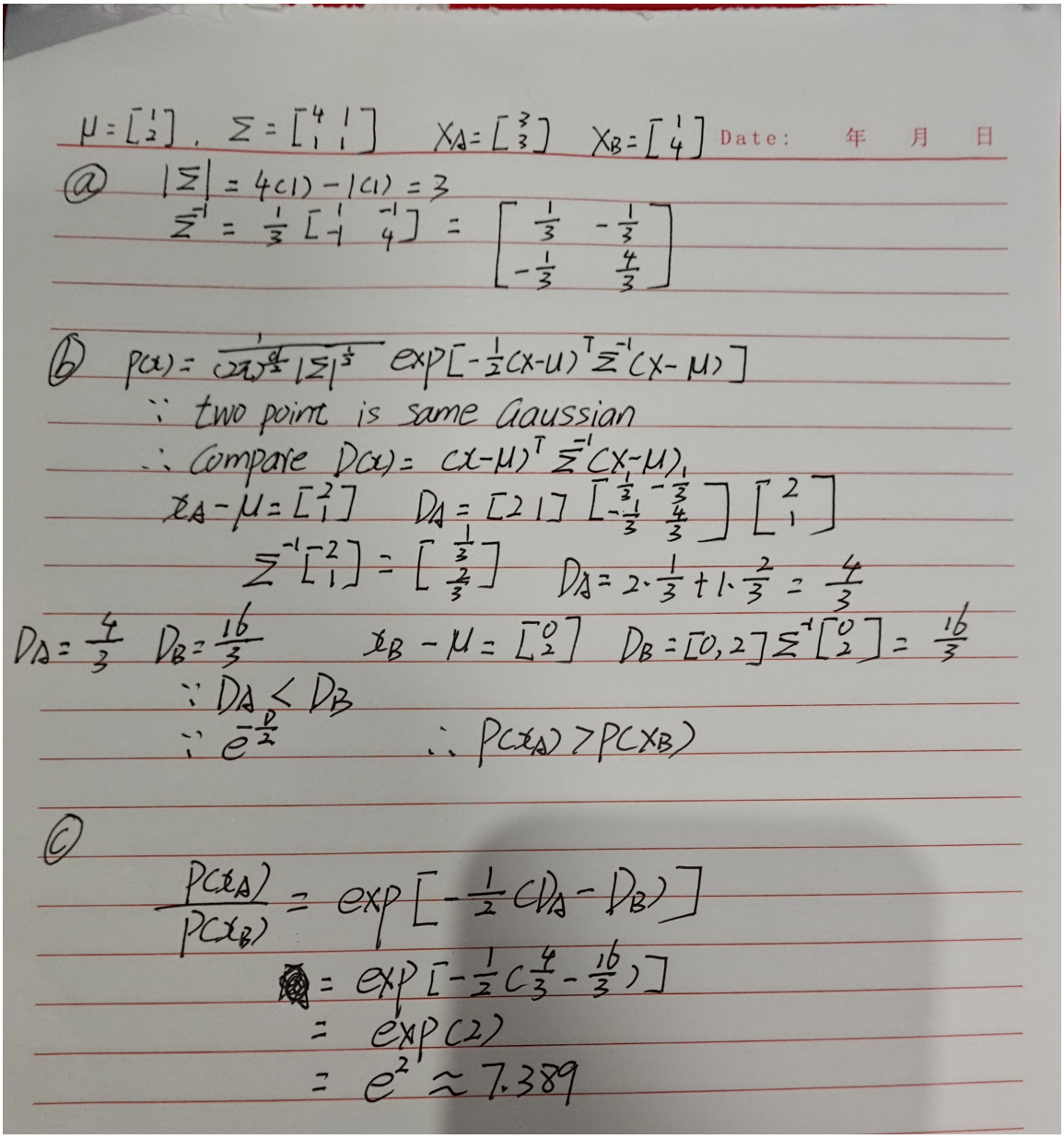
(a) Compute the inverse covariance matrix. [2 marks]

(b) Determine which observation has the larger Gaussian density under the model. Your calculation must use the geometry of the multivariate Gaussian rather than Euclidean distance alone. [5 marks]

(c) Compute

$$\frac{p(\boldsymbol{x}_A)}{p(\boldsymbol{x}_B)}.$$

You may leave the answer in exponential form. [3 marks]



Question 2: Maximum-Likelihood Linear Regression [10 marks]

Four observations are

$$(-1, 0), \qquad (0, 1), \qquad (1, 2), \qquad (2, 2),$$

where each pair is (x_i, y_i).

Assume

$$y_i = w_0 + w_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

where w_0, w_1, and σ^2 are all unknown.

(a) Find the maximum-likelihood estimates of w_0 and w_1. Show the system of equations you solve. [5 marks]

(b) Using your fitted regression model, find the maximum-likelihood estimate of σ^2. [3 marks]

(c) Compute the predicted response at $x_* = 3$. [2 marks]

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(a)

$$p(y_i|x_i, w_0, w_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - w_0 - w_1 x_i)^2}{2\sigma^2}\right]$$

$$\sum_i (y_i - w_0 - w_1 x_i)^2 \qquad \sum_i (y_i - w_0 - w_1 x_i) = 0$$

$$\sum_i x_i (y_i - w_0 - w_1 x_i) = 0$$

$n = 4$, $\sum x_i = 2$

$\sum x_i^2 = (-1)^2 + 0^2 + 1^2 + 2^2 = 6$ $\sum y_i = 5$

$\sum x_i y_i = (-1)(0) + 0(1) + 1(2) + 2(2) = 6$

$4w_0 + 2w_1 = 5$ $2w_0 + 6w_1 = 6$

$\therefore w_0 = 0.9$ $w_1 = 0.7$ $\hat{y} = 0.9 + 0.7x$

(b)

$$\sum_i (y_i - \hat{y}_i)^2 = (-0.2)^2 + (0.1)^2 + (0.4)^2 + (-0.3)^2$$

$$= 0.04 + 0.01 + 0.16 + 0.09 = 0.3$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_i (y_i - \hat{y}_i)^2 = \frac{0.3}{4} = 0.075$$

(c)

$\hat{y} = 0.9 + 0.7x$ $\hat{y} = 0.9 + 0.7 \cdot 3 = 3$

$\hat{y}^* = 3$

Question 3: Bayesian Inference over Discrete Models [10 marks]

An unknown Bernoulli success probability π is known to take one of only three possible values:

$$\pi \in \{0.2, 0.5, 0.8\}.$$

Before observing data, the prior probabilities are

$$P(\pi = 0.2) = 0.2, \qquad P(\pi = 0.5) = 0.5, \qquad P(\pi = 0.8) = 0.3.$$

You then observe the four outcomes

$$1,\ 1,\ 1,\ 0.$$

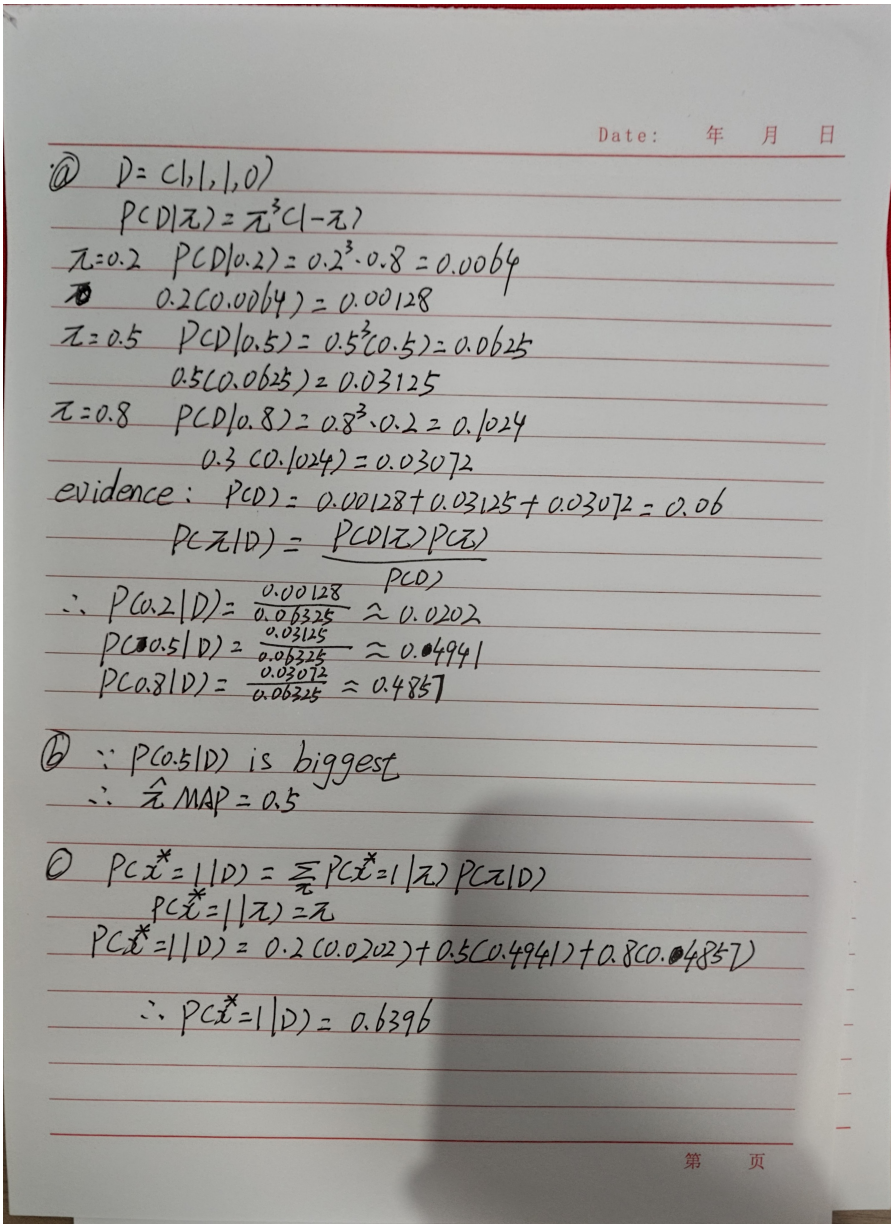
(a) Compute the posterior probability of each possible value of π. [5 marks]

(b) Determine the MAP estimate of π. [2 marks]

(c) Using the full posterior distribution rather than the MAP estimate, compute the posterior predictive probability

$$P(x_* = 1 \mid \mathcal{D})$$

for the next Bernoulli trial. [3 marks]



Question 4: One Complete EM Iteration [20 marks]

Consider the observations

$$x_1 = 0, \qquad x_2 = 2, \qquad x_3 = 4.$$

A two-component one-dimensional Gaussian mixture is initialized with

$$\pi_1^{(0)} = 0.6, \qquad \pi_2^{(0)} = 0.4,$$

$$\mu_1^{(0)} = 0, \qquad \mu_2^{(0)} = 4,$$

and

$$(\sigma_1^2)^{(0)} = (\sigma_2^2)^{(0)} = 2.$$

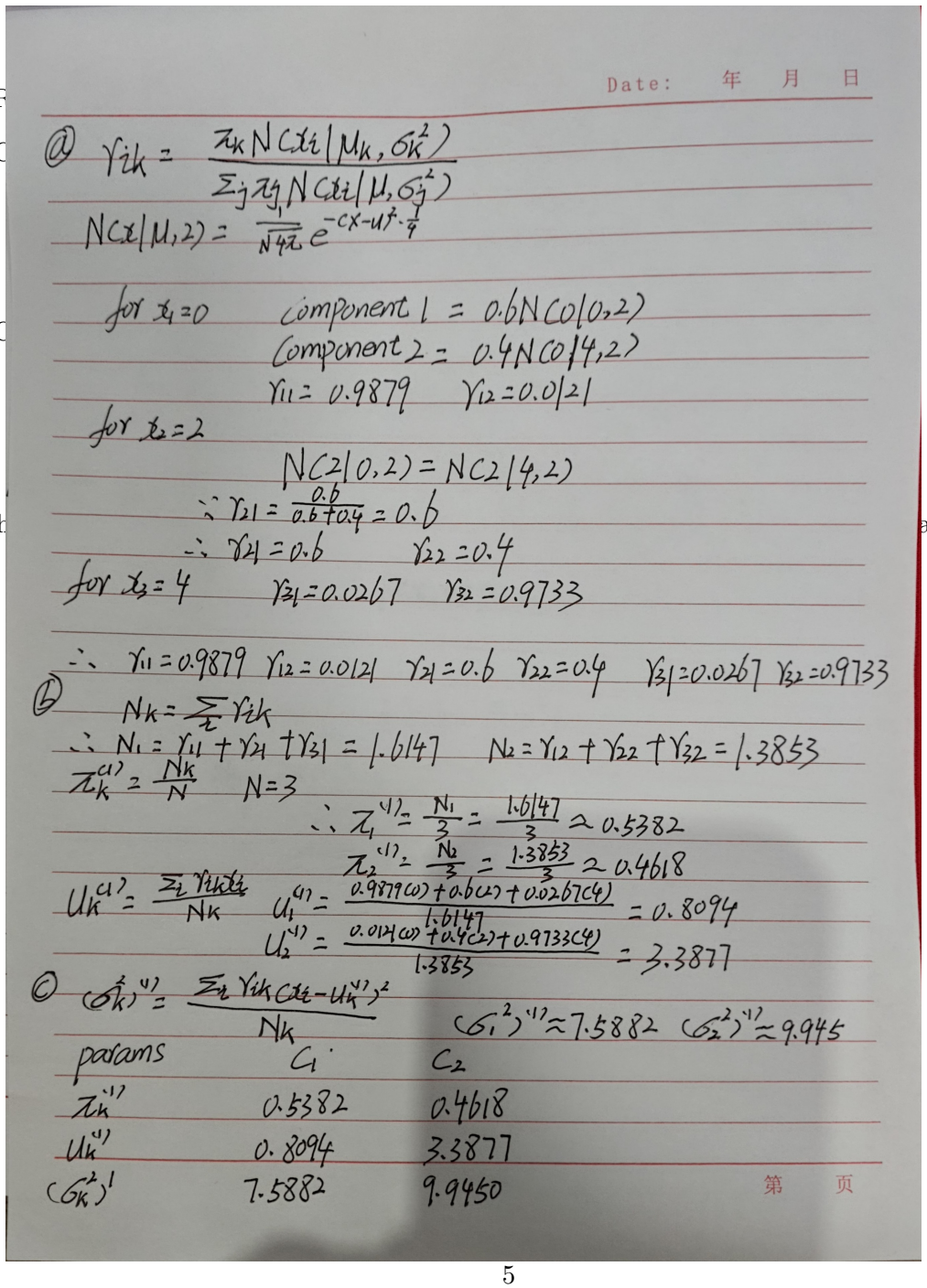
Perform one complete EM iteration. Do not assume hard cluster assignments.

(a) Carry out the E-step and compute all responsibilities [8 marks]

(b) C [7 marks]

(c) C [5 marks]

Use th are provided.



Question 5: Variance of a Linear Projection [10 marks]

Let

$$\boldsymbol{X} \in \mathbb{R}^d, \qquad \mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\boldsymbol{X}) = \boldsymbol{\Sigma}.$$

For fixed $\boldsymbol{a} \in \mathbb{R}^d$ and $b \in \mathbb{R}$, define

$$Y = \boldsymbol{a}^\top \boldsymbol{X} + b.$$

Starting from the definitions of expectation and variance:

(a) Derive the mean of Y. [3 marks]

(b) Derive the variance of Y and express the final result using $\boldsymbol{a}$ and $\boldsymbol{\Sigma}$. [7 marks]

5.

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(a) $E[Y] = E[a^T x + b]$ $E[Y] = a^T E[x] + b$

$E[Y] = a^T \mu + b$

(b) $Var(Y) = Var(a^T x)$ $Var(Y) = E[(Y - E[Y])^2]$

$Y - E[Y] = a^T (x - \mu)$

$Var(Y) = E[(a^T (x-\mu))^2]$

$(a^T v)^2 = a^T v v^T a$

$Var(Y) = E[a^T (x-\mu)(x-\mu)^T a]$

$= a^T E[(x-\mu)(x-\mu)^T] a$

$E[(x-\mu)(x-\mu)^T] = \Sigma$

$\therefore Var(Y) = a^T \Sigma a$

Question 6: MLE with Unequal Observation Variances [10 marks]

Suppose

$$x_i \sim \mathcal{N}(\mu, \sigma_i^2), \qquad i = 1, \ldots, n,$$

independently, where the variances

$$\sigma_1^2, \ldots, \sigma_n^2$$

are known but need not be equal. The common mean μ is unknown.

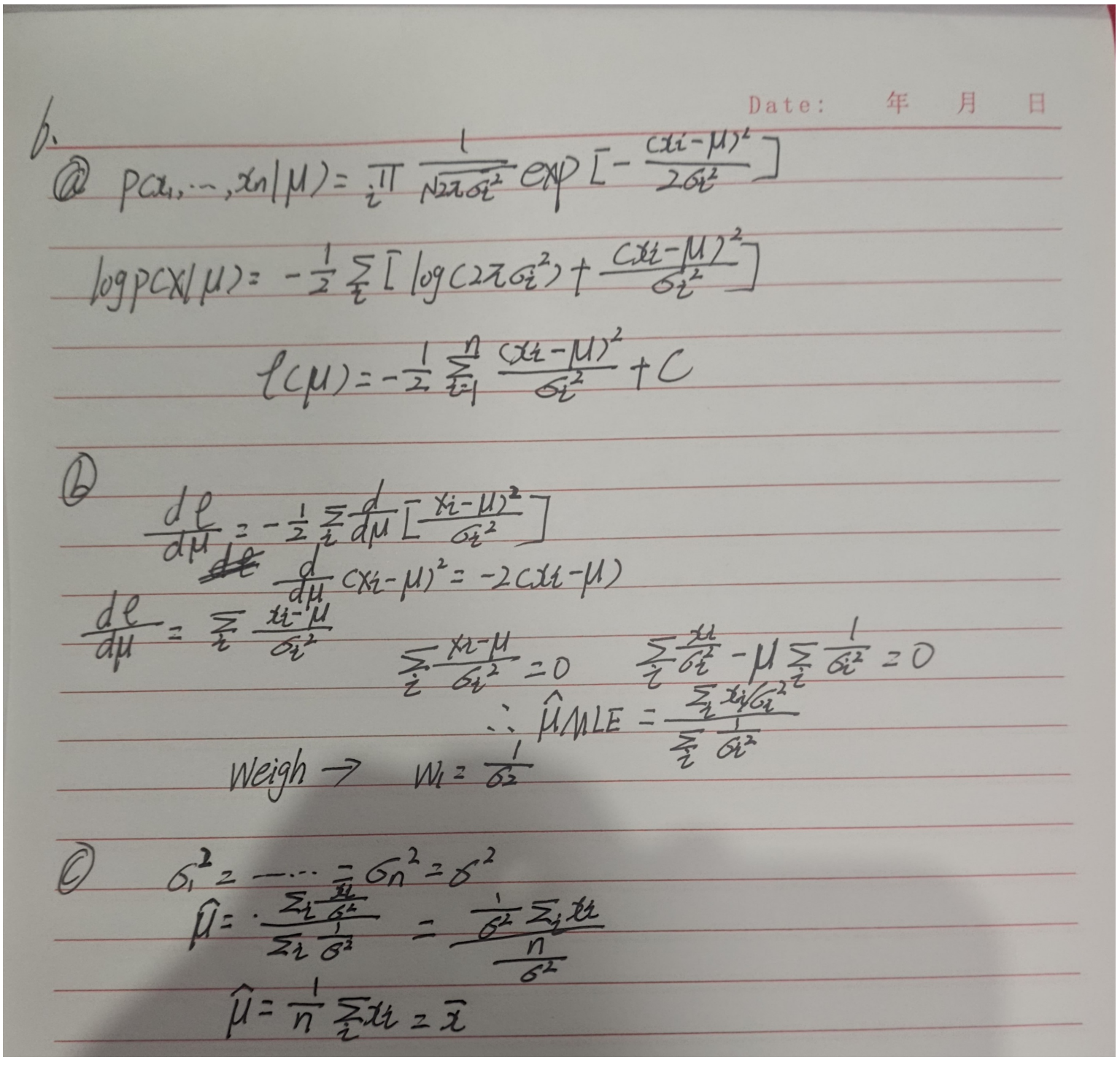
(a) Write the log-likelihood of μ up to terms that do not depend on μ. [3 marks]

(b) Differentiate the log-likelihood and derive the closed-form maximum-likelihood estimator of μ. [5 marks]

(c) Simplify your result when

$$\sigma_1^2 = \cdots = \sigma_n^2.$$

[2 marks]



Question 7: MAP Estimation with a Laplace Prior [10 marks]

Suppose

$$x_i \mid w \overset{\text{iid}}{\sim} \mathcal{N}(w, \sigma^2), \qquad i = 1, \ldots, n,$$

where σ^2 is known.

Instead of a Gaussian prior, assume

$$p(w) \propto \exp(-\lambda |w|), \qquad \lambda > 0.$$

Let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

(a) Derive the negative log-posterior objective for w, ignoring terms independent of w. [3 marks]

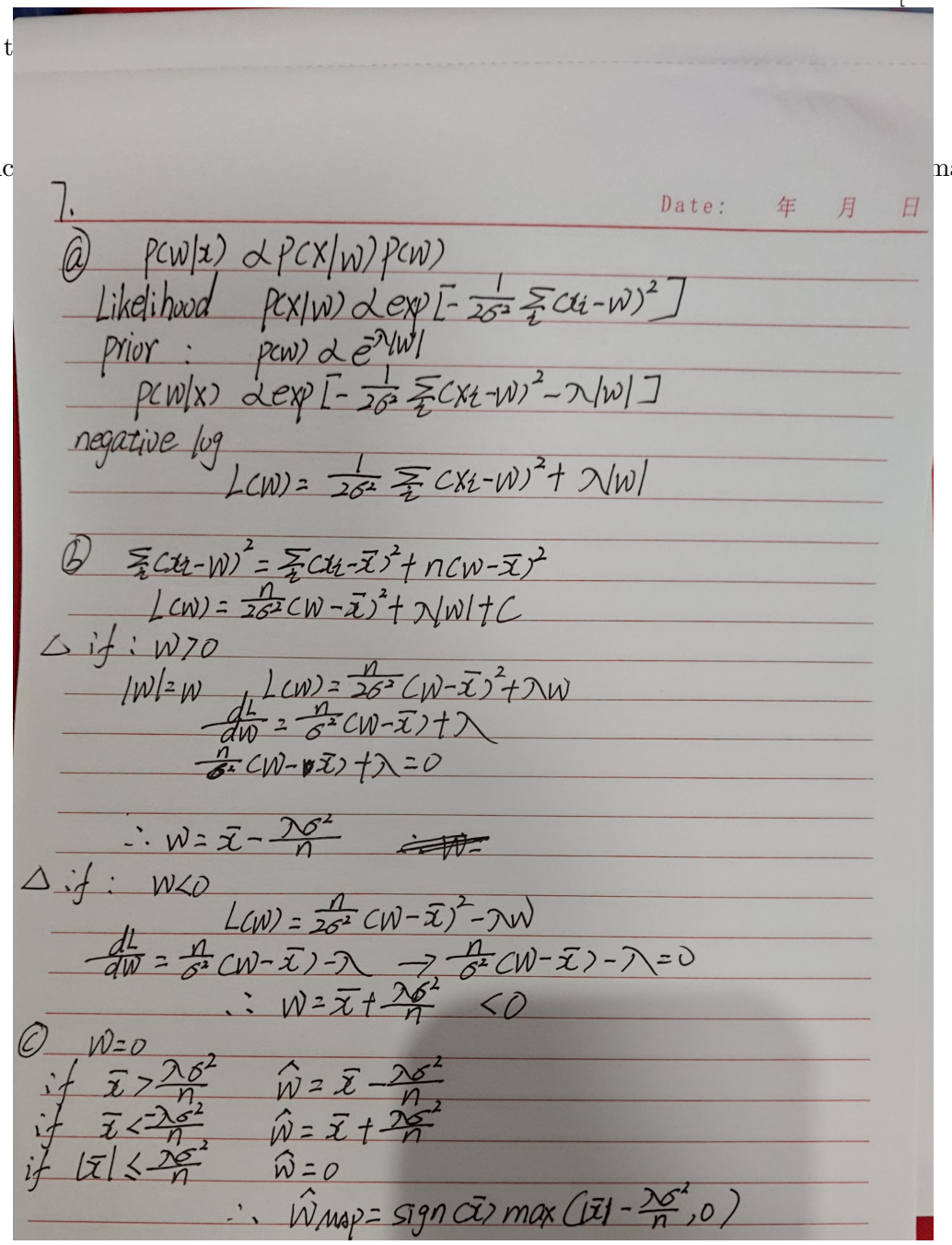
(b) Solve the MAP optimization separately for the cases

$$w > 0 \qquad \text{and} \qquad w < 0.$$

[4 marks]

(c) Include t

as a func marks]



Question 8: From Gaussian Mixtures to K-Means [20 marks]

Consider a Gaussian mixture model with K components satisfying

$$\pi_k = \frac{1}{K}, \qquad \boldsymbol{\Sigma}_k = \sigma^2 \boldsymbol{I}$$

for every component k.

The component means are

$$\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K.$$

(a) Starting from Bayes' rule and the Gaussian density, derive an expression for the responsibility

$$\gamma_{ik} = P(z_i = k \mid \boldsymbol{x}_i)$$

in terms of the squared Euclidean distances

$$\|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|_2^2.$$

[5 marks]

(b) For two components a and b, derive the ratio

$$\frac{\gamma_{ia}}{\gamma_{ib}}$$

in terms of

$$\|\boldsymbol{x}_i - \boldsymbol{\mu}_a\|_2^2 \qquad \text{and} \qquad \|\boldsymbol{x}_i - \boldsymbol{\mu}_b\|_2^2.$$

[5 marks]

(c) Assume component k^* is the unique nearest centroid to $\boldsymbol{x}_i$. Using your result from part (b), determine the limits of

$$\gamma_{ik^*} \qquad \text{and} \qquad \gamma_{ij},\ j \neq k^*,$$

as

$$\sigma^2 \to 0.$$

[5 marks]

(d) Now suppose these limiting hard assignments are written as

$$r_{ik} \in \{0, 1\}.$$

Starting from

$$J_k(\boldsymbol{\mu}_k) = \sum_i r_{ik} \|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|_2^2,$$

derive the centroid value that minimizes J_k. [5 marks]

8.

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(a) bayes $r_{ik} = P(z_i = k | x_i) = \frac{P(x_i | z_i = k) P(z_i = k)}{\sum_j P(x_i | z_j = j) P(z_i = j)}$

$\because \pi_k = \frac{1}{K}$

$\therefore r_{ik} = \frac{\frac{1}{K} N(x_i | \mu_k, \sigma^2 I)}{\sum_j \frac{1}{K} N(x_i | \mu_j, \sigma^2 I)}$

$N(x_i | \mu_k, \sigma^2 I) = \frac{1}{(2\pi\sigma^2)^{\frac{D}{2}}} \exp\left[-\frac{\|x_i - \mu_k\|_2^2}{2\sigma^2}\right]$

$r_{ik} = \frac{\exp\left(-\frac{\|x_i - \mu_k\|^2}{2\sigma^2}\right)}{\sum_j \exp\left(-\frac{\|x_i - \mu_j\|^2}{2\sigma^2}\right)} \approx$ Softmax

(b)

$\frac{r_{ia}}{r_{ib}} = \frac{\exp\left[-\frac{\|x_i - \mu_a\|^2}{2\sigma^2}\right]}{\exp\left[-\frac{\|x_i - \mu_b\|^2}{2\sigma^2}\right]}$ $\quad \frac{e^A}{e^B} = e^{A-B}$

$\frac{r_{ia}}{r_{ib}} = \exp\left[\frac{\|x_i - \mu_b\|^2 - \|x_i - \mu_a\|^2}{2\sigma^2}\right]$

(c) $\sigma^2 \to 0$ $\|x_i - \mu_{k^*}\|^2 < \|x_i - \mu_j\|^2$, $j \neq k$

$\therefore \|x_i - \mu_j\|^2 - \|x_i - \mu_{k^*}\|^2 > 0$

if $\sigma^2 \to 0$

$\therefore \frac{r_{ia}}{r_{ib}} \to +\infty$ $\therefore \frac{r_{ik^*}}{r_{ij}} \to \infty$ $\therefore r_{ik^*} \to 1$

else $r_{ij} \to 0$ $j \neq k^*$

(d) $\frac{\partial J_k}{\partial \mu_k} = 0$ $J_k = \sum_i r_{ik} (x_i - \mu_k)^T (x_i - \mu_k)$

$\frac{\partial J_k}{\partial \mu_k} = -2 \sum_i r_{ik} (x_i - \mu_k)$

$\sum_i r_{ik} x_i - \mu_k \sum_i r_{ik} = 0$

$\mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}$

$\therefore \mu_k^* = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}$